

Literature Review: An Expert-Guided Multimodal AI Ecosystem for Diagnostic Intelligence

CS-405: Deep Learning (Assignment 3)

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1 Abstract

This literature review surveys the current state of deep learning in medical imaging, specifically focusing on the intersection of multimodal brain tumor segmentation, continual learning (CL), and automated report generation via Large Language Models (LLMs). While state-of-the-art models like nnU-Net and Transformers achieve high segmentation accuracy, they remain static, suffering from catastrophic forgetting when protocols change. Furthermore, the translation of segmentation maps into actionable clinical reports remains a disconnected process. We review 15 key publications to identify these gaps, highlighting the need for a unified ecosystem that combines Mixture-of-Modality-Experts (MoME) for segmentation, elastic weight consolidation for adaptability, and grounded LLMs for transparent reporting.

2 Introduction

Brain tumors, such as gliomas, require precise delineation of whole tumor (WT), tumor core (TC), and enhancing tumor (ET) sub-regions for effective treatment planning. Magnetic Resonance Imaging (MRI) is the gold standard for this diagnosis, relying on multiple modalities (T1, T1ce, T2, FLAIR) to capture different tissue properties.

Despite the success of deep learning architectures in automating this process, current clinical deployment faces significant hurdles. First, models trained on specific datasets often fail to generalize to new scanners or patient demographics (distribution shift). Second, updating these models with new data typically degrades performance on previous data, a phenomenon known as catastrophic forgetting. Finally, the output of these models, often just color-coded masks, lacks the semantic richness required for clinical decision making.

This review synthesizes literature across three domains: (1) **Multimodal Segmentation architectures** capable of handling missing data; (2) **Continual Learning** strategies to maintain model longevity; and (3) **Medical LLMs** for generating grounded diagnostic reports.

3 Related Work

3.1 Multimodal MRI Segmentation

Robust segmentation requires architectures that effectively fuse information across MRI modalities.

U-Net and nnU-Net: Ronneberger et al. [1] introduced the U-Net, an encoder-decoder architecture with skip connections that became the de facto standard for medical segmentation. Building on this, Isensee et al. developed nnU-Net [2], a self-configuring framework that automatically adapts hyperparameters to the dataset fingerprint. nnU-

Net remains the strongest baseline in the BraTS challenges, though it lacks specific mechanisms for missing modalities or temporal adaptation.

Transformers in Vision: To capture long-range dependencies that CNNs often miss, Hatamizadeh et al. proposed Swin UNETR [3]. By utilizing a Swin Transformer encoder connected to a CNN decoder, this model demonstrates superior performance in volumetric segmentation tasks, particularly for tumors with irregular shapes.

Missing Modality Robustness: Clinical scans often lack specific sequences (e.g., missing T1ce). Zhang et al. introduced mmFormer [4], which uses a transformer-based cross-modal attention mechanism to hallucinate features from missing modalities, ensuring robust segmentation even when inputs are incomplete. Similarly, Zhou et al. proposed nnFormer [5], interleaving convolution and self-attention to stabilize training on smaller medical datasets.

Relevance: While nnU-Net and Swin UNETR provide strong backbones, our project focuses on the robustness seen in mmFormer. We aim to extend these architectures using a Mixture-of-Experts (MoE) approach to handle distinct modalities more efficiently.

3.2 Continual Learning in Medical Imaging

Medical data streams are non-stationary; scanners and protocols change over time. CL aims to learn sequentially without forgetting.

Elastic Weight Consolidation (EWC): Kirkpatrick et al. [6] pioneered EWC, a regularization-based method that penalizes changes to network weights crucial for previous tasks. While effective for simple classification, Baweja et al. [7] demonstrated its limitations in high-dimensional segmentation spaces, suggesting that EWC alone is often insufficient for complex MRI tasks.

Replay Strategies: Shin et al. proposed Deep Generative Replay [8], where a separate GAN generates fake samples of past data to mix with new training data. This avoids privacy issues associated with storing real patient data. However, GAN training is computationally unstable.

Dynamic Architectures: Perkonigg et al. introduced Dynamic Memory [9], a method specifically for medical imaging that maintains a diverse subset of past images. By selecting samples that cover the distribution of unseen domains, they achieved significantly higher retention rates than standard rehearsal methods.

Recent Surveys: A comprehensive survey by Ozgun et al. [10] categorizes medical CL into regularization, rehearsal, and architectural methods. They conclude that hybrid approaches, combining small episodic memories with regularization, currently offer the best trade-off between privacy and performance.

Relevance: Our project will implement a hybrid CL strategy, utilizing EWC for weight protection and a small, privacy-compliant replay buffer inspired by Perkonigg et al., ensuring the segmentation model adapts to new scanners without degrading.

3.3 Medical LLMs and Report Generation

The final bridge to clinical utility is transforming visual data into text.

Foundational Medical LLMs: Singhal et al. introduced Med-PaLM [11], an LLM fine-tuned on medical exams and consumer queries. It set a new benchmark for clinical reasoning but is computationally prohibitive for local deployment. Conversely, Han et al. developed MedAlpaca [12], a smaller, open-source model fine-tuned on medical instructions, demonstrating that smaller models can achieve high proficiency if data quality is high.

Multimodal Integration: Liu et al. proposed LLaVA-Med [13], adapting the LLaVA architecture to biomedicine. By training on image-text pairs from PubMed, it allows users to query images conversationally. However, these models often suffer from "hallucination," inventing details not present in the image.

Grounded Reporting: To combat hallucinations, Thirunavukarasu et al. [14] emphasize the need for "grounding", linking text generation to structured data. Methods like ChatCAD [15] utilize intermediate structured outputs (e.g., CAD findings) to prompt LLMs, ensuring the generated report aligns strictly with visual evidence.

Relevance: Our project adopts the ChatCAD/MedAlpaca philosophy. We will not feed images directly to the LLM; instead, we will generate a structured JSON from the segmentation mask (Volume, Location) and use that to prompt a fine-tuned MedAlpaca, ensuring zero hallucinations.

4 Gaps and Opportunities

4.1 Identified Gaps

1. **Static Deployment:** State-of-the-art segmentation models (nnU-Net, Swin UNETR) are static. They require full retraining when new domains appear, which is computationally expensive and impractical for hospitals.
2. **Disjoint Pipelines:** Current research treats segmentation and reporting as separate tasks. There is a lack of unified frameworks that automatically convert segmentation metrics into natural language reports.
3. **Hallucination Risks:** End-to-end Vision-Language Models (like LLaVA-Med) are prone to hallucinating tumors or regions that do not exist, making them unsafe for unsupervised clinical use.

4.2 Proposed Contribution

To address these gaps, our project proposes an **Expert-Guided Multimodal AI Ecosystem**. We aim to:

- **Unify Segmentation and Reporting:** Create a pipeline where segmentation maps are deterministically converted into JSON evidence, which then drives the LLM generation. This "Hard Grounding" solves the hallucination issue found in LLaVA-Med.
- **Enable Lifelong Learning:** Integrate Elastic Weight Consolidation (EWC) and Replay Memory into a Mixture-of-Modality-Experts backbone. This allows the model to incrementally learn new MRI protocols without catastrophic forgetting.
- **Enhance Interpretability:** Provide a user interface that visualizes the "link" between the generated text sentence and the specific image region (atlas location), fostering clinician trust.

5 Architecture

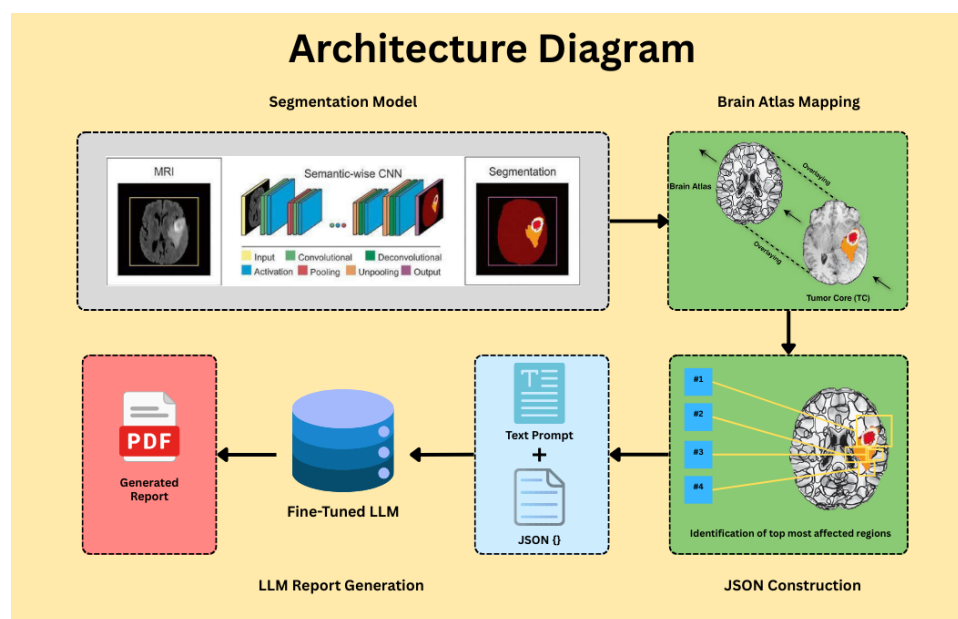


Figure 1: Architecture

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